

Van Le

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Quantitative analyst with a B.A. in Computational and Applied Mathematics and applied experience in time-series forecasting, ML model evaluation, and unsupervised learning at scale. Recent work: walk-forward cross-validation on 10M+ equity records, validation of 8 mechanistic models against 570K+ data points at Merck, and PCA/clustering pipelines on 100K+ single-cell RNA-seq samples.

CORE SKILLS

Programming: Python (pandas, numpy, scikit-learn, PyTorch, SciPy, seaborn), SQL, R, Bash, Scala

Statistical & Machine Learning: Time-Series Forecasting, Walk-Forward Cross-Validation, A/B Testing, Regression, Classification, Clustering (PCA, k-means, DBSCAN), Random Forest, XGBoost, Model Evaluation Methodology

Data Engineering & Cloud: ETL Pipeline Design, Data Modeling, Data Warehousing, AWS (S3, EMR, Athena), PySpark, MySQL

BI & Visualization: Tableau, Excel, Plotly

PROFESSIONAL EXPERIENCE

Data Analyst **M&T Tax Firm** **Oct 2024 - Present**

- Cut reconciliation cycle time from 3 - 5 days to 1 day with 95% fewer reporting errors by designing optimized SQL workflows on point-of-sale and banking transaction data, replacing manual spreadsheet processes.
- Automated monthly financial statement generation and data validation across client portfolios, saving 15+ hours/month, by building 3 reusable SQL query templates with embedded data quality checks.
- Conducted YoY variance analysis on revenue growth, expense ratios, and cash flow trends across 10+ client portfolios, surfacing \$175K in cost-reduction and tax-benefit opportunities adopted into client operational strategies.

Data Science Co-op **Merck & Co.** **May 2023 - Aug 2023**

- Contributed to 7% yield increase and 25% shorter production cycles by validating 8 mechanistic models against 570K+ experimental data points using statistical model evaluation methodology; findings adopted by process engineers into manufacturing changes.
- Reduced data ingestion time by 40% for 3 - 4 downstream analytics and engineering teams by building performant AWS EMR-to-S3 ETL pipelines for manufacturing process and instrument data.
- Designed domain-aware schemas and reusable SQL and Python templates on AWS Athena for data retrieval and batch-vs-standard analytics, replacing manual folder searches and cutting time-to-data by 60% across 3 - 4 teams.

Data Science Intern **Adaptive Biotechnologies** **Jun 2022 - Aug 2022**

- Initiated a new flow cytometry research direction by developing unsupervised ML pipelines (PCA, k-means, DBSCAN) on 100K+ single-cell RNA sequencing samples, surfacing novel T-cell receptor clustering patterns presented to department leadership.
- Improved experiment-tracking efficiency by 13% for 2 life-sciences research teams by gathering requirements from end users and consolidating fragmented manual reporting into a unified Tableau BI layer with real-time visibility into experiment status, sample throughput, and result tracking.

PROJECTS

Stock Recommendation System: End-to-End ML Platform for Equity Discovery [\[GitHub\]](#) **PySpark, scikit-learn, Streamlit, Plotly**

- Architected a PySpark ETL pipeline ingesting 10M+ OHLCV records across 3,600+ NASDAQ tickers, joining exchange metadata into a unified analytical schema enriched with 20+ engineered indicators; applied PCA to compress dimensionality into 5 principal components, yielding a 386K-row query-ready dataset that reduced downstream latency and removed feature multicollinearity.
- Selected LinearSVR via walk-forward time-series CV from 5 benchmarked models (ARIMA, Random Forest, SVR, XGBoost); chose MAPE as primary metric since R-squared is misleading on price series, methodology transferable to production model evaluation.
- Built an interactive Streamlit dashboard with Plotly visualizations integrating the validated forecasting model with PCA-compressed features, exposing ticker-level recommendations and price predictions through a no-code query interface.

Budget Efficiency Model for Marketplace Campaign Optimization [\[GitHub\]](#) **SQL, R, DuckDB, tidymodels, A/B Testing, Shiny**

- Built an end-to-end statistical evaluation and ML pipeline to identify low-efficiency marketplace campaign segments, surfacing \$20.3K in spend consuming 34.6% of budget while generating only 22.9% of approved conversions.
- Designed offline experiment logic to simulate model-guided budget reallocation, projecting 148 incremental approved conversions and reducing CPA from \$58.53 to \$50.98 under fixed-budget constraints.

Chronic Disease Risk Intelligence: Predicting Outbreak Patterns Across the U.S [\[GitHub\]](#) **Scikit-learn, Random Forest, Tableau**

- Trained a Random Forest model on 1.7M CDC patient records to identify high-risk individuals and designed Tableau dashboards to visualize population-level risk distribution, enabling data-driven public health prioritization.

EDUCATION

M.S. in Computational Data Analytics (Online), *Georgia Institute of Technology* **Aug 2026 - Expected December 2028**

B.A. in Computational and Applied Mathematics, *The State University of New York at Buffalo* **Aug 2022 - Dec 2023**

Associate of Arts in Business and Chemistry, *North Seattle College* **Jan 2018 - Jun 2020**